

AI-Assisted Federal Survey Harmonization

Research Fact Sheet

2026-02-10

What's the problem?

This research examines **47 Census Bureau demographic survey instruments** containing approximately **7,000 questions**, each designed for a specific analytical purpose. No complete map exists showing where these surveys overlap, where their questions are compatible, and where the differences matter. Building that map manually represents months of effort for a skilled research team. This research accomplished that work in weeks using AI-assisted methods, at a total API cost under \$100.

Why should leadership care?

Response rates are declining across surveys. Every year, it costs more to collect less. Before looking to alternative data sources, we should be in-sourcing: getting more value from data we already collect.

The Census Bureau already *has* enormous data that could answer questions no single survey can. Cross-survey integration — using overlapping questions as bridge variables — increases the explanatory power of existing collections without adding respondent burden. Where questions turn out to be interchangeable, the same analysis identifies consolidation opportunities and encourages language standardization across instruments, improving data quality and fit-for-purpose alignment.

How is this different?

Traditional survey harmonization is manual, slow, and limited to 2–3 surveys at a time. This research uses **multiple AI models from different vendors independently rating question pairs**, avoiding single-model bias. The approach holds all 47 surveys in view simultaneously, finding connections no individual expert could surface — not because experts lack skill, but because no one holds 7,000 questions in working memory at once.

Does it work?

Agreement between independent models from competing vendors is measured, not assumed: post-arbitration agreement on harmonization judgments reached $\kappa = 0.84$, and a confirmation run one model generation later agreed with the original analysis on which pairs are harmonization candidates for 88% to 91% of shared pairs ($\kappa = 0.70$ to 0.72). Results are not driven by any single model's bias. Among the questions that had ACS territory to be compared against, the rate at which a harmonization path is found is similar for both surveys (54.8% for CPS, 47.5% for FoodAPS), suggesting the rate reflects shared conceptual territory rather than a survey-specific artifact. The framework passes basic sanity checks: demographics like race and age score as near-100% consolidable, while genuinely different constructs are correctly flagged with specific barrier diagnoses. The full pipeline, data, and code are public and reproducible.

Key findings

Across two major survey pairs (CPS and FoodAPS vs. ACS), 275 unique source questions had enough shared subtopic territory with ACS to be compared, and **142 of those 275 (~52%)** have at least one viable harmonization path. Each question is assigned its best tier across all of its ACS pairs:

- **Tier 1 – Direct recode** (51 questions, ~19%): Interchangeable with simple recoding. High-confidence bridge variables.
- **Tier 2 – Statistical adjustment** (91 questions, ~33%): Linkable with documented transformations. Usable bridges with known limitations.
- **Tier 3 – Not linkable** (133 questions, ~48%): Different constructs. The framework identifies these and explains why.

Measured against each full instrument rather than the compared subset, questions with a harmonization path are 86 of 211 CPS questions (~41%) and 56 of 462 FoodAPS questions (~12%). The gap reflects how much of each survey's content shares no subtopic territory with ACS at all.

These results give survey managers a scored inventory of where cross-survey integration adds value and where instruments can be aligned for better return on collection investment.

What's at stake?

Without this work, 47 surveys continue as isolated silos. Declining response rates erode quality with no compensating strategy. Connecting existing surveys increases the overall value of current collections, helps each survey collect the data it's best positioned to gather, and builds resilience as the data environment shifts. The methodology extends naturally to surveys across the broader federal statistical system.

What's next?

Stage 4 moves from pairwise comparisons to the full multi-survey network, discovering multi-hop integration paths (Survey A to B to C) across all 47 Census demographic surveys.

DRAFT — These results are preliminary and have not been independently validated or peer reviewed. Full methodology, data, and code: github.com/brockwebb/federal-survey-concept-mapper. The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.